Can one reading rule account for the structure of the Standard Model? A compatibility map with registered checks

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

Version 2.52, 2026-10-15. Archived on Zenodo, https://doi.org/10.5281/zenodo.23404761. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Student version. This PDF is the student version of the record, written with an AI assistant from the archived text. The archived text remains on Zenodo: https://doi.org/10.5281/zenodo.23404761

Abstract

The structure of the Standard Model can be written with known constructions. Gauge fields are transport matrices on lattice links (Wilson). The electroweak model has a Higgs doublet (Glashow, Weinberg, Salam). The charge pattern of one family is that of Pati–Salam and of Furey's number-operator construction. A single light lattice fermion with exact chiral symmetry comes from the Ginsparg–Wilson relation and Neuberger's overlap operator. This record asks whether one reading rule from the author's framework, Mathematical Materiality (MM), gives these pieces a common meaning. It goes from the data to MM: for each piece, the measured fact, the reading, whether they are compatible, the published construction that already does the same, and what would make the reading incompatible. Selected gauge-sector structures and the charge pattern of one family admit a common interpretation under stated additional hypotheses. The checks concern separate constructions, and their joint chiral gauge realisation remains to be established. No mathematical priority is claimed, and the checks provide no experimental discrimination from the Standard Model. Files labelled "registered predictions" were fingerprinted with SHA-256 before their corresponding calculations; subsequent rechecks are identified separately, and predictions not met are reported. The fingerprints were recorded locally and are not publicly time-stamped.

The forces and particles of nature as readings of one structure: a map of what fits and what does not

Student version of MM record 2.52 (DOI 10.5281/zenodo.23404761), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Version 2.52 of the Mathematical Materiality (MM) series, 15 October 2026. This work continues three earlier records:

- MM 2.40, the Feynman checkerboard (DOI 10.5281/zenodo.23280168);
- MM 2.27, the photon as a phase on the lattice link (DOI 10.5281/zenodo.22667516);
- MM 2.38, anomaly-free charges with a right-handed neutrino (DOI 10.5281/zenodo.23055978).

A correction to know right away. This work also reports a counting error in MM 2.40. It is discussed in section 7.

1. What this work is, and what it is not

The question

The **Standard Model** is the theory that describes the particles and the forces between them, except gravity. It works very well. But it has many different pieces: different forces, different charges, colours, masses. The question of this work is: can a single **reading rule** give a common meaning to the *structure* of the Standard Model?

By structure we mean five things:

- which forces there are;
- which charges the particles have;
- how many "colours" there are;
- why the charged weak current acts on only one "hand" of the particles;
- where the masses come from.

The work goes from the data to MM, never the other way. For each piece of the Standard Model it states, in this order:

- the measured fact;
- the MM reading;
- whether the two are compatible, that is, whether they agree;
- the construction already published by others that does the same thing;
- what would make the reading incompatible.

The result

Some selected structures of the gauge sector of the Standard Model admit a common MM interpretation. The same holds for the charge pattern of one family of particles. But this holds under some additional hypotheses, listed in section 3. The pieces that fit are four:

- three colours, with eight **gluon** modes that interact with each other (gluons carry the strong force);
- the four bosons of the electroweak force, with a massless photon and $\rho = 1$ at tree level (we explain this in section 4.5);
- fractional charges, such as 1/3 and 2/3, with a neutral neutrino and a neutral neutron;
- one single species of light Dirac particle on the lattice. From it a single "hand" (a Weyl sector) is selected with a projection. In the U(1) backgrounds tested, its calibrated index agrees with the topological charge.

We will explain these words one at a time. For now it is enough to know that they are four separate checks. The checks concern separate constructions. Putting them together in a single chiral gauge theory remains to be established.

The parameters are not read. The parameters are the numbers that the theory does not fix by itself. Here they are taken from the data or left open:

- the weak mixing angle θ_W;
- the scale v;
- the masses;
- the quark and neutrino mixings;
- the strong CP angle $\bar{\theta}$.

What it is not

- **It is not a derivation of the Standard Model.** Every construction used here is already known:
- Wilson's lattice gauge theory;
- the Glashow–Weinberg–Salam model;
- the Pati–Salam charge pattern and Furey's number-operator construction;
- the Ginsparg–Wilson relation and Neuberger's overlap operator;
- Kaplan's domain wall.

Section 2 lists them before anything else. **No mathematical priority is claimed:** the work does not say that it discovered this mathematics first.

- What is specific to MM is only the reading rule. It is the rule that gives these constructions a common meaning. Together with the rule there are the registered checks: where the rule fits the data and where it does not.
- These checks do not make it possible to tell MM apart from the Standard Model with an experiment.

How the statements are kept apart

The text keeps four kinds of statements apart.

- **Published results** by others (section 2).
- **Readings.** They are the author's premises translated into physical terms. What the translation adds to the author's words is marked as an addition of the translator (section 3).

- **Numerical checks.** Some predictions are "registered". This means that they were written in a file and fixed with a fingerprint before the calculation. In this way they cannot be changed after seeing the result. The checks made afterwards are labelled as such. The registered predictions that did not come true are also reported (section 6).
- **Review.** Each working round, and this package, was checked by an independent artificial intelligence system. Its corrections are applied here (section 7).

2. Other people's work first

First of all, here are the already published results that this work uses. Many words in the table will be clearer later. The important point is this: every piece of mathematics used here was already known.

topic	published result used here
lattice fermion doubling	Nielsen, Ninomiya, Nucl. Phys. B 185, 20 (1981)†; for discrete time (Floquet): Sun et al., arXiv:1806.09296 (2018), where the theorem still holds; Higashikawa, Nakagawa, Ueda, arXiv:1806.06868 (2018), one Weyl node in a band from a map with winding W = 1
random lattices	Griffin, Kieu, hep-lat/9211022 and hep-lat/9307011: with a gauge-invariant coupling the doublers come back
Ginsparg–Wilson relation and overlap	Ginsparg, Wilson, Phys. Rev. D 25, 2649 (1982)†; Neuberger, hep-lat/9707022 and hep-lat/9710089 (1998); Hernández, Jansen, Lüscher, hep-lat/9808010 (1999), locality
domain wall fermions	Kaplan, hep-lat/9206013 (1992); Shamir, hep-lat/9303005 (1993); Golterman, Jansen, Petcher, Vink, hep-lat/9309015 and hep-lat/9312005 (1993)
chiral gauge theories on the lattice	Lüscher, hep-th/0102028 (2001): abelian chiral theories constructed non-perturbatively, non-abelian ones to all orders of perturbation theory. Kaplan, Les Houches lectures, arXiv:0912.2560 (2009): no practical way is known to put general non-abelian chiral theories on the lattice. Kadoh, Kikukawa, arXiv:0709.3658 (2008): a non-perturbative construction of the Glashow–Weinberg–Salam model, gauge-invariant, covering all SU(2) topological sectors with zero U(1) magnetic flux
lattice gauge fields	Wilson, Phys. Rev. D 10, 2445 (1974)†: transport matrices on the links, plaquette action
electroweak model	Glashow (1961), Weinberg (1967), Salam (1968)†; PDG 2025, review 10 "Electroweak model and constraints on new physics"
charges from a number operator	Furey, arXiv:1603.04078 (2016) and arXiv:1405.4601 (2014)
the Pati–Salam charge pattern	Pati, Salam, Phys. Rev. D 10, 275 (1974)†, the lepton as a fourth colour; SO(10) unification (PDG 2025, review "Grand Unified Theories")
anomaly-free charges with ν_R	MM 2.38 (DOI 10.5281/zenodo.23055978): with a right-handed neutrino and Dirac masses the anomalies leave the hypercharge of the quark doublet free
quantum cellular automata	D'Ariano, Perinotti, arXiv:1306.1934 (2014); Bisio, D'Ariano, Tosini, arXiv:1212.2839 (2015); MM 2.40
Higgs self-coupling at the Planck scale	Shaposhnikov, Wetterich, arXiv:0912.0208 (2010): $m_H \approx 126$ GeV from $\lambda(M_P) \approx 0$ under their assumptions for high energies; Buttazzo et al. (2013), as used in MM 2.38
mass hierarchy from a fifth direction	Arkani-Hamed, Schmaltz, hep-ph/9903417 (1999)
winding of a product of maps	an identity for the winding of a product, derived in the working notes and reproduced in section 4.7; in the literature it is associated with the name Polyakov–Wiegmann (not checked at the source here)

† Standard references cited from the reviews listed. They were not re-read at the source for this work.

Data. The data come from PDG 2025 (S. Navas et al., Phys. Rev. D 110, 030001 (2024) and 2025 update). The PDG is the official collection of measured values in particle physics. The reviews used concern: the electroweak model, the Higgs boson, the number of light neutrino types, the muon decay parameters, the CKM matrix, neutrino mixing, conservation laws. The summary tables of mesons and baryons are also used. Other data sources:

- Bagdonaite et al., arXiv:1311.3438 (2013): Δμ/μ from methanol at z = 0.89;
- Burns, Keus, Sher, Tait, arXiv:2402.08626 (2024), and Yoo, Scherrer, astro-ph/0211545 (2003): v at the time of nucleosynthesis;
- Uzan, arXiv:1009.5514 (2011): a review of the constants that might change over time.

3. The reading rule, in physical terms

The **premises** are the author's starting points, like the axioms in geometry. On 15 October 2026 there were ten of them. Others come from earlier: from the MM book and from 27 September 2026. Here they are stated in physical terms. The author's exact words, in Italian, are only in the original registrations. There is also an informal account of the picture behind them, with its words explained. But nothing in the checks depends on that account.

The rule in one sentence

Particles and forces are not different objects. They are different readings of the same structure.

To understand, imagine a **lattice**: a grid of points in space, like the crossings on a sheet of squared paper. Between two neighbouring points there is a **link**, like the side of a square. Each reading assigns three things:

- to each link of the lattice, a **transport** of the readings involved. The transport says how a particle changes when it passes from one point to the next. For light it is a phase, that is, an angle of rotation, like the position of a clock's pointer. For the weak force it is two units. For colour it is three readings;
- to each **fermion**, that is, to each particle of matter, two things: the number of readings it shares, which is its charge; and the hand it is read with, which is its **chirality**;
- to the **Higgs field**, the projection that links the two hands. This projection is the mass.

Chirality is like the difference between the right hand and the left hand. They are equal but mirror images: they cannot be superimposed. Particles too have a "left-handed" version and a "right-handed" version.

Its parts

The rule has six parts, from R1 to R6. In the third column there is what the translation into physical terms adds to the author's words.

	reading	added by the translation
R1	**Three internal readings.** Inside a fermion there are three binary readings, that is, of the yes/no kind (taken or not taken): $2^3 = 8$ states. The colour says which readings are involved. A transport along a link can mix them: it is a unitary 3×3 matrix.	That the three readings are equivalent and that any unitary matrix can mix them. That the transport preserves a chosen complex volume form on C^3. This imposes det U = 1, so SU(3) and not U(3).
R2	**Charge from the shared readings** (rule R-Q). |Q| = N_shared/3. N_shared is 1 or 2 for colour triplets (the quarks) and 0 or 3 for singlets (the leptons). Conjugate states have opposite charges.	The split between triplets and singlets is an added rule. It already contains the distinction between quarks and leptons.
R3	**Three families.** The same triple of readings, seen from the other side, gives the three families.	That the number of colours is equal to the number of families. Only the number is read, not the masses.
R4	**Light and the weak force.** Light is a phase on the link (MM 2.27). The weak force is the same transport, with two units mixed on the link: one full and one slightly smaller. It is a $U(2) = [SU(2) \times U(1)]/Z_2$ matrix.	That the "slightly smaller unit" is the second state of the doublet.
R5	**Mass and the Higgs.** The mass of a fermion is the projection that links its two hands. The Higgs field is where this projection happens, with one scale v. The Higgs mass comes from the self-coupling of that field. The low-temperature electroweak scale is a fixed structural parameter.	That "self-resonance" is the quartic coupling λ. That "history" is read as structure (reading L8-a) and not as elapsed time (reading L8-b).

R6	**One hand as an act of reading.** For charged fermions, the left-handed and the right-handed component form a Dirac field, after electroweak symmetry breaking. The charged weak current acts on the left-handed component. Light acts on both. The reading places the difference in the projector, not in the field. On the lattice the light species has a twin. The twin is pushed to the scale of the cell when the reading level lies above the bottom of the "hill" of the lattice spectrum and below its first ridge.	The two components have different electroweak representations, and the Yukawa coupling remains part of the construction: the projector alone does not assign them. The window $0 < m_0 < 2r$ is a condition of the construction, not a consequence of the premises. The author said "the top of the hill is passed"; reading it as "the bottom of the valley is passed" is a new interpretation of the metaphor, given by the translator. Whether neutrinos are Dirac or Majorana remains open.

A few words for the table. A **matrix** is a table of numbers that transforms another list of numbers. **Unitary** means that it preserves lengths, like a rotation. **SU(3), SU(2), U(1)** are the names of the symmetry groups of the three forces: strong, weak and electromagnetic. **det U = 1** is a condition on the determinant of the matrix: it is what separates SU(3) from U(3). A **triplet** is a group of three states that transform together, like the three colours of a quark. A **singlet** is a state on its own. The **doublet** is a pair, like electron and neutrino for the weak force.

4. The map, piece by piece

Now let us look at the pieces of the Standard Model one at a time. For each piece there are the same entries: the data, the reading, the check, whether it is compatible, who had already done it, what is not read. The labels in brackets (such as P2a or G1) are the names of the registered checks. They help to find them again in the table of section 6.

4.1 The charges of one family (registered checks P2a–P2c)

- **Data.** The charges of the particles are 0, ±1/3, ±2/3, ±1. The neutron and the hydrogen atom are neutral. For example the u quark has charge 2/3, the d quark −1/3, the electron −1.
- **Reading.** R1 + R2, that is, rule R-Q: the charge counts the shared readings, divided by 3.
- **Check.** The forces of the theory must be **anomaly-free**. Here an anomaly is a defect that appears when going from classical to quantum physics: if it hits the symmetries of the forces, the theory is no longer consistent. Let us take the anomaly-free family with a right-handed neutrino ν_R and Dirac masses (MM 2.38). There the charges depend on a free number: the **hypercharge** Y_Q of the quark doublet. The formulas are: q_u = Y_Q + 1/2, q_d = Y_Q − 1/2, q_ν = 1/2 − 3Y_Q, q_e = −1/2 − 3Y_Q. The four anomalies vanish for every value of Y_Q. Rule R-Q instead admits exactly two values: Y_Q ∈ {−1/6, +1/6}. To find them, all fractions with denominator up to 60, between −3 and 3, were tried. In addition there is an exact solution.
- Y_Q = +1/6 is the Standard Model: neutrino 0, electron −1, u 2/3, d −1/3, neutron 0, hydrogen 0.
- Y_Q = −1/6 is the same content, with two changes. One takes Q → −Q. And one exchanges the two members of every doublet: u ↔ d, ν ↔ e, with their right-handed partners, the Higgs with its conjugate and the Yukawa couplings. The colour does not change. This was checked afterwards. **The registered form of the relabelling (Q → −Q with 3 ↔ 3̄) was wrong.**
- Without ν_R the rule adds nothing. The anomalies already fix Y_Q = 1/6, and the rule admits it.
- **Compatible: yes.** With ν_R the rule fixes the charge where the anomalies alone leave it free.
- **Place already taken.** Pati–Salam and SO(10) also fix the charges with ν_R. And Furey (2016) obtains the charge pattern from a number operator, on the same space of 8 states. R-Q is not a derivation of those models. It is an added rule that contains the distinction between quarks and leptons.
- **Not read.** The value of the fine-structure constant, that is, the strength of the electromagnetic force. The Yukawa couplings.

4.2 The algebra of the three readings (registered check T-A)

The 8 states of R1 can be described with three switches, each on or off. Mathematically one uses three operators ε_i (i = 1, 2, 3) that turn a switch on or off. The space of all the states is called a three-mode Fock space. It is the exterior algebra of C^3. It is the same construction as in phase 2.29 of the MM 2.27 deposit and in Furey (2016). Here are the five checks.

- **A1.** The operators obey the basic rules of fermions (canonical anticommutators). Exact: error 0.
- **A2.** The possible values of the number operator N are {0, 1, 1, 1, 2, 2, 2, 3}. So N/3 gives 0, 1/3 (three times), 2/3 (three times), 1. These are exactly the charges. A technical note. In the archived convention, Σ ε_i†ε_i counts the empty readings (3 − Σb), and N_shared = Σ ε_i ε_i†. The reviewer noted that in the log the labels of the sectors N = 1 and N = 2 are swapped with respect to the operator. The spectrum and the Casimir values do not change.
- **A3.** There are nine combinations ε_i†ε_j. They commute with N: that is, they do not change the charge. Their traceless part generates su(3), the algebra of colour. The **Casimir** is a number that characterises a group of states. It is 4/3 on the sectors with one and two readings (two conjugate triplets) and 0 on the other two.
- **A4.** The spin generators of phase 2.29 are S_i = −(i/2) C_j C_k. They do not commute with N (‖[S_i, N]‖ = 1). Stated conditionally: these spin generators cannot act on the internal index with this charge. A representation is needed in which spin and charges commute. Usually separate factors are used.
- **A5.** The fifteen products C_iC_j, K_iK_j, C_iK_j conserve the parity of N, that is, whether N is even or odd.

They generate an algebra of dimension 15. It is irreducible on the even and odd sectors (1 + 3 and 3 + 1). **Correction** (by the reviewer, rechecked). These raw products generate so(3,3) ≅ sl(4,R), a non-compact form. With the Hermitian generators (C_i, iK_i) the algebra is instead the compact su(4), of dimension 15, with 4 → 3 + 1. This is the Pati–Salam pattern: the lepton as a fourth colour. It is not the full Pati–Salam model. The weak factors, the identification of the physical generators and the dynamics are missing.

4.3 Three colours, three families (registered check T-B; data only, low weight)

- **Data.** One measures how many light neutrinos couple to the Z particle: N_ν = 2.9963 ± 0.0074. It is 0.5σ from 3. The symbol σ is the standard deviation: 0.5σ means very close. There is also R_ℓ = Γ(hadrons)/Γ(leptons) at the Z. It is the ratio between how many times the Z decays into hadrons and how many into leptons. It is roughly proportional to the number of colours. With 2 or 4 colours it would change by about 33 %.
- **Reading.** R3: the same triple read from two sides.
- **Compatible: yes, with low weight, as registered.** The number three appears everywhere in MM: three directions, eight states = 2^3, rank 3 of the book's decomposition (which holds when the prime supports are disjoint). So a rule that gives 3 is not surprising. The figure 1/3 in the registration is an example with an assumed distribution. It is not a significance. Then N_ν counts only the light active neutrinos coupled to the Z. It does not count all possible families.

4.4 The gluons (registered checks G1–G3)

- **Data.** There are eight massless gluons that interact with each other. There is no ninth long-range force coupled to baryon number (baryon number counts protons and neutrons).
- **Reading.** The photon method of MM 2.27 is used, that is, a phase on the link. But with the transport of the three readings (R1). On each link there is a 3×3 matrix, U = exp(iA). The **plaquette action** S = Σ_p Re Tr(1 − U_p) is used. A plaquette is the small square of the lattice: one goes around its four sides. Static fields, that is, fields at rest, are used on a periodic 6^3 lattice: a cube of 6 points per side that closes on itself.
- **Check.**
- **G1.** Three wavevectors are tried: (1,0,0), (1,1,0), (1,2,3). A wavevector says in which direction a wave goes and how closely packed it is. In all three the static quadratic form has 16 complex transverse amplitudes for su(3). That is 8 × 2: eight gluons, each with two modes. (The program prints half the rank of the real sine/cosine form: 32.)
- **G3.** With u(3) instead of su(3) one finds 18. The ninth gluon appears only without det = 1.
- **G2.** One looks for a cubic term in S − S_2, for a generic su(3) configuration whose pieces do not commute. The cubic term is the sign that the gluons interact with each other. One finds $c_3 = -39.0$ from a fit $c_3 + c_4\varepsilon$, with ε down to 1e-4 (that is, 0.0001). This recheck was done afterwards. For fields that commute the cubic term vanishes. The compact plaquette action instead keeps higher even powers. **The registered criterion was not met.** It asked for a ratio constant within 5 % between ε = 1e-2 and 1e-3. But at ε = 1e-2 the quartic term weighed as much as the cubic one.
- **Compatible: yes.** But be careful: these are transverse amplitudes of the static quadratic form. They are not verified counts of propagating polarisations. And c_3 is a coefficient of that configuration. det = 1 is the added hypothesis of R1. The central U(1) of U(3) is not automatically baryon number.
- **Place already taken.** Wilson (1974): with matrices on the links, Yang–Mills theory follows by construction. The MM part is only the meaning of the link (three readings) and the reason offered for det = 1. **Confinement** (quarks are never seen alone) and the running of α_s are not redone here.

4.5 The electroweak bosons: two units on the link (registered checks E1–E4)

- **Data.** There are four bosons: W^+, W^-, Z and γ (the photon). The W have charge ±e. The triple and quartic couplings between bosons were measured at the LEP 2 accelerator. They are those of a non-abelian theory, that is, one where the order of the operations matters. M_W = 80.369 GeV, M_Z = 91.188 GeV. The photon has no mass. Then ρ = 1 at lowest order. The number ρ compares the masses of the W and the Z with the weak angle: ρ = 1 means that they are linked in the simplest way.
- **Reading.** R4 + R5. On the link there is a U(2) transport. And there is a Higgs doublet frozen at its vacuum (0, v/√2), in unitary gauge. Unitary gauge is a convenient way to choose the internal coordinates.
- **Check.**
- **E1.** In each of the three tested wavevectors there are 8 complex transverse amplitudes: 4 bosons × 2.
- **E2.** The registered statement on the cubic term is supported in two ways. First: the calculation with a generic field gives $c_3 = -29.4$. Second: for fields that commute there is no cubic term. For W3 and B only, $(S - S_2)/\varepsilon^3$ goes to zero like ε: −0.77, −0.23, −0.06. The absence is also proved by hand. With fields that commute the plaquette is exp(iεH_p), with H_p Hermitian and independent of ε. The real part of its trace contains only even powers of ε. The original program said "not met". But it used a fit criterion that is not in the registration: an intercept of 0.0144 against a tolerance of 1e-3. A possible explanation of the residual is the loss of numerical

precision in the subtraction. No comparison at higher precision is supplied here.

- **E3.** The registration gives the squares of the physical masses: 0, $g^2v^2/4$ (twice) and $(g^2 + g'^2)v^2/4$. The program built the matrix K of the quadratic form c^T K c. Its eigenvalues are 0, 3201.7 (twice) and 4164.7 GeV². The program compared them directly with the predicted physical values. But for real fields normalised in the standard way M^2_phys = 2K holds. **The original automatic comparison forgot a factor of two**, and so it said "not met". The original program also printed ρ = 2. It came from a wrong formula: in the numerator there was the predicted physical M_W^2, in the denominator the computed eigenvalue K_Z. A common rescaling of the two masses would instead leave ρ unchanged. The subsequent calculation uses consistent normalisations. It recovers the registered values: 0, 6403.44 (twice) and 8329.35 GeV². And ρ = 1.000000000000. It is an identity at **tree level**, that is, at the first level of calculation, the simplest one. The twelve digits describe the numerical check, not an experimental precision. The original program also checked the direction of the photon, W3/B = tan θ, and the charge of the W: +1 in units of the photon coupling. One chooses g = 0.65. This gives M_W = 80.022 GeV and M_Z = 91.265 GeV.
- **E4.** M_W/M_Z = cos θ holds. With the MS-bar value $\sin^2\theta = 0.23122$ it gives 0.8768, against 0.8814 measured (−0.52 %). It was registered in advance as the known difference between two conventions, MS-bar and on-shell. It was not a test of MM.
- **Compatible: yes, for some selected identities of the bosons.** With a U(2) representation of the link the checks reproduce:
- four bosons;
- a cubic self-interaction for fields that do not commute;
- a massless photon;
- massive W and Z, with ρ = 1 at tree level, and W charge ±1.

This is not yet an assembled electroweak model. Two things are missing. The normalisation of the two kinetic terms, which fixes g and g′ independently. And the action on the representations of the fermions.

- **Place already taken.** The Glashow–Weinberg–Salam model on a lattice: with 2×2 matrices on the links and a doublet, all this follows by construction. The MM part is the meaning of the two states.
- **Not read.** θ_W (the ratio g′/g is taken from the data), v, and which hand the weak force acts on. The ratio between the two units might one day fix g′/g at some scale. No such rule was written in advance. And none is sought afterwards.

4.6 One light hand (registered checks O1–O5, H1–H2)

This is the hardest part. Let us go step by step.

- **Data.**
- The charged weak current acts on left-handed fermions. The muon decay parameters say so (PDG 2025, review 57, table 57.1). The 90 % limits are: |g^V_LL| > 0.960, |g^V_RR| < 0.017, |g^V_LR| < 0.023, |g^V_RL| < 0.105. In words: the "left-left" coefficient is large, the ones that involve the right hand are very small.
- These data must be read in a precise framework: four fermions, local interaction, without derivatives, with lepton number conserved. In that framework the measurements of muon decay and of inverse muon decay support the V−A charged interaction of the leptons, within the errors. The helicity −1 of the neutrino ν_μ in pion decay also supports it. These limits concern specific operators and low-energy processes.
- No light "mirror" fermions with the same weak charges are observed. For the charged fermions of the Standard Model, the Z couples both to the left hand and to the right hand. A ν_R that does not feel the electroweak force does not have this coupling.
- **The obstacle.** Putting particles on a lattice has a famous problem: **fermion doubling**. On a lattice every particle carries along some unwanted **twins**, like an echo. The Nielsen–Ninomiya theorem says that, under certain assumptions, this always happens. The construction called **overlap** uses a special form of the symmetry between the two hands: the Ginsparg–Wilson form. Section 4.7 shows what doubling means for the checkerboard of MM 2.40.
- **Reading.** It is R6, implemented as a standard overlap operator. It is built with the matrices and parameters of MM 2.40. The Dirac matrices come from the checkerboard. The Wilson parameter is r = 0.15 and the level is m_0 = 0.2 (the mass of the 2.40 domain wall). Time is treated as a fourth direction of space (Euclidean). In formulas: D_W(k) = i Σ γ_μ sin k_μ + r Σ (1 − cos k_μ), A = m_0 − D_W, D = 1 − A(A†A)^(−1/2). Alongside, the standard set (r = 1, m_0 = 1) is also run, for comparison. **It is not shown** that this Euclidean operator is the limit of the 2.40 tick automaton.
- **Check.**
- **O1.** The Ginsparg–Wilson relation γ5D + Dγ5 = Dγ5D holds to 9e-16 for both sets. This is the machine precision.
- **O2.** In the free field D has a single zero, at k = 0. At the other 15 corners D = 2. A zero corresponds to a light particle. So there is only one light particle, and the 15 twins are heavy. Outside |k| < 0.3, the smallest singular value on a 16^4 grid is 1.056 (for 2.40) and 0.390 (standard).

- **O3.** With $\gamma\tilde{5} = \gamma 5(1 - D)$, the Weyl operator $P_+ D \hat{P}_-$ has a cone at $k = 0$ and a gap elsewhere. The cone has slope $1/m_0 = 5$ for 2.40. A **gap** is a jump in energy: it means that there are no light particles there.
- **O4, locality.** One looks at how much the operator "feels" distant points. The norm of the kernel $f(r)$ at Manhattan distance r decreases. The Manhattan distance is counted as in a city of blocks: only along the streets, not diagonally. The fitted effective slopes are 0.4054 over distances 4–14 (2.40) and 0.8049 over distances 2–8 (standard). Both calculations used 48^4 sites instead of the registered 32^4. The registered range for 2.40 was 0.1–0.4. **So O4 was not met for 2.40, by a small margin.** For the standard set the numerical criterion is met, on the modified lattice.

Be careful: the fitted slopes are not the asymptotic decay rates, that is, at very large distance. The 2.40 kernel has an axial complex-momentum singularity at $k = (\pi + i\kappa, 0, 0, 0)$, with $\kappa \approx 0.101$. It has a diagonal one at $k = (\pi + i\eta, i\eta, i\eta, i\eta)$, with $\eta \approx 0.050$. The reviewer found them; both were rechecked. So the asymptotic rate in the Manhattan distance used in O4 is not established here. The operator is local with exponential tails, not ultralocal. **The earlier working phrase "the twin fades over 2.5 cells" is withdrawn.**

- **O5, the index with the light field.** The lattice is 6^4, with constant U(1) fluxes ('t Hooft). The **index** counts the difference between left and right zero modes. The **topological charge** Q counts how many times the field "twists". The calibrated index is equal to Q for $Q = 0, 1, 2, -1, 0$, both for 2.40 and for the standard set. The gap of H_W stays open (≥ 0.0999 and ≥ 0.74). For 2.40 the registered prediction was a mismatch at $L = 6$. The reason: the flux per plaquette (0.17) exceeds the margin $2r - m_0 = 0.1$. **The anticipated mismatch was not observed** in the five backgrounds tested. No general conclusion on reliability follows from this.
- **Index convention.** With $H_W = \gamma 5(m_0 - D_W)$ the identity $n_+ - n_- = \frac{1}{2}\,\mathrm{Tr}\,\mathrm{sign}(H_W)$ holds. The first program counts the exact zero modes as $n_- - n_+$. After the first background, a second, faster program uses the spectral asymmetry. The two methods agree on the first background, $Q = 0$, where the sign is not tested. The sign convention is calibrated separately, at $Q = 1$, as registered. So the index is not an independent prediction. The fast method does not exclude extra pairs of zero modes with opposite chirality.
- **H1, the "hill".** One counts the number of light species and their net chirality, as m_0/r varies. The net chirality is the coefficient of the index after the Weyl projection. For $m_0/r = -0.5, 1, 3, 5, 7, 9$ one finds: 0/0, 1/+1, 5/−3, 11/+3, 15/−1, 16/0. This holds for $r = 1$ and for $r = 0.15$.
- **H2.** With $Q = 1$ the index is $1, -3, 3, -1, 0$ for $m_0 = 1, 3, 5, 7, 9$.
- **Compatible: yes, in a window.** Imagine the spectrum of the lattice as a landscape. At $k = 0$ there is the bottom of a valley. Further away there are ridges and peaks. The level m_0 must lie above the bottom of the valley and below the first ridge: $0 < m_0 < 2r$. The 2.40 values ($0.2 < 0.3$) are inside. Above the top ($m_0 > 8r$) all the twins return and the net chirality cancels. In the Ginsparg–Wilson form the projector of a field depends on the operator itself: $\hat{P} = (1 - \gamma 5(1 - D))/2$. So the difference between the hands sits in the reading operator. This is R6 inside a known object. It is a compatibility, not a derivation.
- **Precisely what is shown.** The overlap operator has one light Dirac species, with both chiralities. A Weyl sector, that is, one single hand, is selected with a projection. These checks do not construct the fermion measure of a chiral gauge theory. They do not construct it even for U(1). And here everything is abelian. Constructions of the electroweak model on the lattice exist, non-perturbative, with stated restrictions (Kadoh–Kikukawa 2008). Implementing one of them is outside this work.
- **Place already taken.** Neuberger (1998), Hernández–Jansen–Lüscher (1999), Kaplan (1992), Shamir (1993). It is already known, in the overlap literature, that the number of species and the index depend on m_0.
- **A side check on pion decay.** The pion can decay into electron and neutrino, or into muon and neutrino. Let us take massless neutrinos and V/A couplings equal for all leptons. At tree level the ratio $\Gamma(\pi \to e\nu)/\Gamma(\pi \to \mu\nu)$ is $1.2833 \cdot 10^{-4}$. It does not depend on the mixture of the hands. The measured value, $1.230 \cdot 10^{-4}$, is consistent with this helicity suppression. The radiative corrections of the Standard Model account for a reduction of roughly four percent. They are not computed here. A pion–lepton coupling equal for all leptons, without derivatives and without an additional lepton-mass factor, would give 5.487. The ratio does not select a hand. **An earlier working note read the ratio as evidence for one hand. That reading is corrected here.** The evidence that the charged current is left-handed is the muon decay data, seen above.
- **Would be incompatible if:**
- an established right-handed charged-current contribution were found, incompatible with the specified chiral interaction;
- a light mirror sector with the stated charges were found;
- the non-locality of the reading operator could be observed at long range.

4.7 The obstacle on the 2.40 checkerboard: winding (registered checks T1–T6)

The 2.40 checkerboard is an **automaton**: at each tick each particle takes one step, along x, then y, then z, in turn. For the part with only one hand (the Weyl sector) this gives a map $U: T^3 \to SU(2)$. The **winding number** W of a map says how many times the map "wraps around" its target. Think of a rubber band wound around a finger: once, twice, no times. The formula is $W = -(1/24\pi^2)\int \varepsilon_{ijk}\,\mathrm{Tr}[R_i R_j R_k]\, d^3k$, with $R_i = U^\dagger \partial_i U$.

For SU(2), when ±I are regular values, W is equal to the sum of the node chiralities. The nodes are the points where light particles appear. The chiralities are defined from the local effective Hamiltonian. They are counted at quasi-energy 0 and, separately, at π, with a consistent sign convention. The registered definition used the sign of the Jacobian of the vector part u of $U = u_0 I - i\, u\cdot\sigma$. It agrees at quasi-energy 0. But at π it has the opposite sign, because there the local effective field is −u (the reviewer noted this).

- **T2.** W = 0 for MM 2.40 (−2e-16 on 48^3 and 96^3 grids).
- **T3.** W = 0 for 200 random automata made of single-direction steps in SU(2). The largest |W| is 1.2e-14, that is, zero. This is also proved: each step is a loop in SU(2), and $\pi_1(SU(2)) = 0$. This means that every loop in SU(2) can be shrunk to a point.
- **T3b.** The Weyl automaton of D'Ariano–Perinotti is exactly a product of three single-direction steps: A± = e^{−iσ_z q_z} e^{±iσ_y q_y} e^{−iσ_x q_x}. So W = 0.
- **T4 (control).** The model of Higashikawa and colleagues gives W = +1.00000, with the sign convention that reproduces their dispersion. W = 1 is met. **The unrestricted registered statement "a single gapless point" is not met.** Uniqueness holds at quasi-energy zero. But degeneracies also appear at π, on the faces of the zone.
- **T5a.** Partial steps: one component of the spin moves, the other waits. With three steps: W = 0 in 64 cases out of 64.
- **T5b and T5c were not met.** The registered prediction was some $W \neq 0$ with 4–8 partial steps. But all 236 cases had W = 0. On the doubled cell the model of Higashikawa gives +1 − 1 = 0, not 2. Both facts are explained afterwards by an identity: $\mathrm{Tr}[((gh)^{-1}d(gh))^3] = \mathrm{Tr}(g^{-1}dg)^3 + \mathrm{Tr}(h^{-1}dh)^3 - 3\, d\,\mathrm{Tr}(g^{-1}dg \wedge dh\, h^{-1})$. So W(gh) = W(g) + W(h) for regular single-valued maps on the torus. In words: the winding of a product is the sum of the windings.
- **Consequence.** Let us take a finite product of steps. Each step is periodic on the same torus and depends on only one component of k. The "coins", that is, the mixing rules, are constant. Then W = 0 in U(n). The conditions matter. The coins must be constant (or have a known W). Each factor must be periodic, not only the product. Higashikawa's half steps are not single-valued on the cell.
- **What it means.** In the checkerboard with separate steps, the Weyl sector does not realise an isolated hand in the bulk. To obtain one single hand one must state which further conditions or sectors enter. Section 4.6 does this with the reading level. **The earlier working phrase "the opposite hand can only be moved" is withdrawn.** Corners, bands and walls are different mechanisms, each with its own assumptions.
- **T1c.** With the Wilson term r = 0.15 the registration expected one light species and 15 heavy species with gap ≥ 0.1. The single light species is confirmed. The other valleys have a gap ≥ 0.160 (grid 40^3). **But the valleys are 64, not 16** (section 7). The original search for minima was incomplete.
- **T6, dispersion.** The dispersion says how the energy of a wave depends on its wavelength. The registration concerned the leading order of the group-velocity deviation. The program instead computed the relative deviation of the energy. The leading orders are supported. The coefficients were corrected by hand afterwards:
- with one tick (2.40 without the palindrome, and D'Ariano–Perinotti) the relative energy shift is linear in κ, with coefficient n_x n_y n_z and opposite sign for the two helicities;
- with the palindrome and with the 2.27 photon there are only quadratic shifts. For the photon, with $k = \kappa n$, the relative deviation of the energy is $-(\kappa^2/24)\,\Sigma\, n_i^4$. The radial deviation of the group velocity is $-(\kappa^2/8)\,\Sigma\, n_i^4$. The registration had written 1/24 for the velocity (checked afterwards).

The registration also pointed to some observable consequences: a linear term at the Planck scale, to be compared with the limits on Lorentz violation. These consequences are not verified here. The comparison would require fixing in advance the operator, the sector and the physical step.

4.8 The Higgs sector (readings compared with the data; no new code)

The **Higgs boson** is the particle linked to the field that gives mass to the other particles.

- **Couplings and mass.** The measured Higgs couplings are consistent with the Standard Model pattern (PDG 2025). At tree level, the couplings with the fermions go as m_f/v. Those with W and Z go as m_V^2/v. The couplings with the photon and the gluons pass through loops, that is, through virtual particles. R5 reads the mass of the fermion and its coupling to the Higgs as the same projection seen twice. **Compatible.** Two limits apply:
- "one scale for all masses" holds only in the electroweak sector. The mass of the nucleon comes from the strong force (QCD);
- "no right hand, no mass" holds for Dirac masses. Majorana terms also exist.
- **The Higgs mass as self-coupling.** In the Standard Model $m_H^2 = 2\lambda v^2$, with λ = 0.129 at first order. The trilinear coupling of the Higgs with itself is measured at $-1.2 < \kappa_3 < 7.2$ (ATLAS, 95 %). The Standard Model gives 1. **Compatible by structure.** Today's data do not allow a distinction. A κ_3 different from 1 would not by itself contradict the reading.
- **λ = 0 at the Planck scale.** It is an additional condition, not contained in the premises. From MM 2.38: $\lambda(M_P) = -0.0090 \pm 0.0052$ with the pole top mass (1.7σ from 0). Or -0.0100 ± 0.0031 with the direct top mass (3.2σ). The place is taken: Shaposhnikov–Wetterich (2010).

- **Numerology declared in advance and not used.** Two coincidences had been noticed before: m_H/v = 0.5085 against 1/2, and λ = 0.1293 against 1/8. They are not used. As an exact relation at first order, m_H = v/2 is excluded: it would give 123.11 GeV against 125.20 ± 0.11 GeV measured.
- **v and the history of the cosmos.** Could v have changed over time?
- Methanol at z = 0.89, that is, 7.5 billion years ago: $\Delta\mu/\mu = (-1.0 \pm 0.8_stat \pm 1.0_sys)\cdot 10^{-7}$, with μ = m_p/m_e (Bagdonaite and colleagues 2013).
- Nucleosynthesis, in the first minutes of the universe: $|\delta v/v| \lesssim 1$ %. It is a bound that depends on the model (Burns and colleagues 2024).
- To go from the bound on μ to a bound on v one needs the relation $\Delta\ln \mu = \Delta\ln m_p - \Delta\ln y_e - \Delta\ln v$, with m_p tied to QCD. So the bounds on a variation of v in the cosmos depend on which dimensionless ratios are compared. And they depend on assumptions about the Yukawa couplings and QCD. They refer to the low-temperature vacuum value, not to the thermal history of the electroweak transition.

The reading L8-a takes the low-temperature electroweak scale as a fixed structural parameter. It is **compatible with these bounds, but it is not the only possible one.** The value $v/M_P \approx 2\cdot 10^{-17}$ (M_P is the non-reduced Planck mass) is not read.

4.9 SU(2) from the records (registered; working result only; no code in this deposit)

A different route was also tried for the group of the weak force. One looks at what happens when a record is transported along a closed history, which returns to the starting point. The set of these loops forms a group, the holonomy group. The record bench of the 2.34 working programme was used. **The registered expectations were not met.**

- With 4 registers: semisimple algebras of dimension 6 (so(4)) or 3 (so(3)), depending on a parameter.
- With 3 registers: abelian (dimension 1) or trivial.
- With 5 registers: undetermined.

Algebra, group and force must be kept apart: so(4) ≅ su(2) ⊕ su(2); SO(4) ≅ [SU(2) × SU(2)]/Z_2; SO(3) is not SU(2). **No stable SU(2) is read from the records today.** The programs depend on unpublished working files of the 2.34 programme and are not included. The registrations and the outcome note are included.

5. What is read, what is not, and what would contradict

This table sums up the map. For each piece it says what the reading gives, what it does not give, and which observation would put it in trouble.

piece	read (structure)	not read or not constructed	observation incompatible with the rule or with its realisation
charges	0, ±1/3, ±2/3, ±1; neutral ν and neutron; Y_Q = ±1/6 with ν_R (4.1)	α, Yukawa couplings	charges or colours that violate R-Q within its stated scope; a non-zero charge of the neutrino or of the neutron, under the assumptions of section 4.1
colour	SU(3) transport, 8 modes of the static form that interact with each other (4.2, 4.4)	confinement, running of α_s (not redone)	evidence that the colour transport requires an additional central U(1) generator, coupled to the same internal states
colours and families	3 and 3 (4.3, low weight)	why the families have different masses	an additional fermion family, or a number of colours different from three (the check on the width of the Z tests only the light active neutrinos)
electroweak bosons	γ, W±, Z; cubic self-interaction; ρ = 1 at tree level; W charge ±1 (4.5)	θ_W, v, M_W, M_Z; the kinetic normalisation; the action on the fermions	evidence that the boson masses cannot be described with a single doublet, as here (additional scalar representations do not necessarily lead to ρ ≠ 1)
one hand	one light Dirac species, a Weyl sector by projection, calibrated index = Q in the tested U(1) backgrounds, in the window $0 < m_0 < 2r$ (4.6)	an anomaly-free chiral fermion measure and the full electroweak coupling	an established right-handed charged-current contribution incompatible with the specified chiral interaction; a light mirror sector with the stated charges; a non-locality observable at long range

Higgs	tree-level couplings with the fermions m_f/v and with W/Z m_V^2/v, with photon and gluons through loops; one electroweak scale; m_H from the self-coupling (4.8)	v, v/M_P, the hierarchy, κ_3	measurements of the Higgs couplings incompatible with the single-doublet pattern used here, including its radiative corrections
λ at the Planck scale (additional condition)	$\lambda(M_P) \approx 0$ (4.8)	—	$\lambda(M_P)$ clearly different from 0 with a better measurement of the top mass
v in time	a fixed low-temperature electroweak scale (4.8)	the value of v	an established variation of a dimensionless ratio that contains v, incompatible with the fixed scale under the stated assumptions on Yukawa and QCD
masses	—	all the fermion masses (taken from the data)	—
mixings	—	CKM, PMNS, CP phases	—
strong CP	—	$\bar{\theta}$, constrained indirectly by the limits on the electric dipole moment to the order of 10^{-10}	—
neutrinos	—	the mass mechanism: the premises do not select it	— (observing neutrinoless double beta decay would establish lepton-number violation; not observing it would not by itself establish that neutrinos are Dirac)
weak group from the records	— (depends on the bench, 4.9)	SU(2) from the records	—

6. Registered predictions and outcomes

This table lists all the predictions registered before the calculations. It says which ones came true and which did not. The predictions that did not come true are also reported, as such.

block	registered	met	not met as registered
T-A three readings	A1–A5	A1–A4; A5 in dimension and irreducibility	identification of A5 (so(3,3) for the raw products, corrected)
T-B colours and families	one comparison	yes (low weight, as registered)	—
T-D wall states of 2.40	D1, D2	D2	D1 (see section 7)
Tappa 1 winding	T1a–T6	T1a (in substance), T2, T3, T3b, T4 (W = 1), T5a, T6 (leading orders)	T1b (64 valleys, not 16); species count of T1c (the single light species holds); T4 unrestricted single gapless point; T5b; T5c
Tappa 2 charges	P2a–P2c	P2a, P2c	P2b (form of the relabelling)
Tappa 3 records	G1–G6	G2	G1, G3, G4, G5 and G6 (n = 3); n = 5 undetermined
gluons	G1–G3	G1, G3	G2 (criterion; the cubic term is there in the recheck)
electroweak	E1–E4	E1 (at the three tested wavevectors), E2, E3 (recovered with consistent normalisation), E4 (as registered, not a test)	— (the original program said "not met" for E2, with an unregistered fit criterion, and for E3, because of a normalisation error in its automatic comparison)
overlap	O1–O5	O1, O2, O3, O4 (standard, on 48^4), O5 (standard)	O4 for 2.40 (0.405 against 0.1–0.4); the anticipated mismatch of O5 for 2.40 (not observed)
hill	H1, H2	H1, H2	—

Be careful: in the row "Tappa 3 records" the labels G1–G6 are the checks of section 4.9, not those of the gluons.

7. Corrections

The corrections are important. They change some earlier readings. This is why they are here, in view.

Corrections to the working notes. The notes remain unchanged in the archive. The later notes and this text prevail.

- The algebra of the fifteen raw products is so(3,3), not su(4). And the labels of the sectors N = 1 and N = 2 were swapped (4.2).
- It was said that the overlap escapes the Nielsen–Ninomiya theorem by giving up strict locality. This reading is corrected. The operator is local with exponential tails. The decisive step is the Ginsparg–Wilson form of the symmetry between the hands.
- The phrases "the twin becomes non-locality" and "the twin fades over 2.5 cells" are withdrawn. The fitted slope is not the asymptotic rate (4.6).
- The phrase "the opposite hand can only be moved" is withdrawn (4.7).
- The pion ratio does not select a hand (4.6).
- The classification of E2 and E3 in the working note is replaced by section 4.5.
- The phrase "exactly the su(4) of Pati–Salam", in the Tappa 2 note, is replaced by section 4.2.

Correction to MM 2.40.

- **The valleys.** The README of MM 2.40 (line 85) says that the massive automaton has 16 species. But the full count over the Brillouin cube $[0, 2\pi)^3$ is different. The Brillouin cube is the set of all the possible wavevectors on the lattice. Without the Wilson term ($\theta = 0.01$), the two-tick palindromic Dirac automaton has **64 Dirac valleys**. They lie at distance θ from their quasi-energy centre: 32 around zero and 32 around $\pi/2$. The exact relation is $\cos \Omega = \cos 2\theta \cdot \cos 2k_x \cdot \cos 2k_y \cdot \cos 2k_z$. The two-tick evolution conserves the eight parity classes of the sites. So there are 8 valleys in each parity sector, each one a Dirac fermion. The earlier check had examined 16 of them. With the Wilson term $r = 0.15$ one single light species remains in the bulk (T1c).
- **The walls.** A domain wall is a boundary where particles can stay trapped. Wall states were searched for in the new valleys (registered check T-D). Extra states were found at half period. But the counting method used does not isolate the wall states. It used the trace of a projector with a window of 0.08. So it also caught bulk states. For this reason D1 is reported as not met. A diagnosis made afterwards gives a first picture:
- with the 2.40 parameters ($\theta_0 = 0.2 > r = 0.15$), part of these states sit on the walls, with a mass ≈ 0.013;
- with the tested value $\theta_0 = 0.1 < r$, the selected states are mostly bulk states.

"One mode per wall" remains valid as a local statement, near $k = 0$ and at quasi-energy 0. A corrected text for MM 2.40 requires a better criterion to say where each state sits. It will be given separately. The programs for this are not part of this deposit.

How the work was done

Author, licences and assistants. Author: Ugo Lanciano, independent researcher. The author is not a professional physicist. The premises are his. The original texts and materials are under the CC BY 4.0 licence, the original code under the MIT licence. The translations into physical terms, the programs and the archived text were made with an AI assistant (Claude), and they are marked as the translator's where they add to the premises. The calculations were made with AI assistants (Claude and Codex). There was no review by a human referee.

Registrations. The "registered" prediction files were written and given a fingerprint (SHA-256, UTC time) before the calculation, in the author's working files. These fingerprints do not have a public time stamp. The exact words of the premises, in Italian, are in the original registrations (`documenti/registrations_original/`). The informal account of the picture behind them is in `WHERE_THIS_COMES_FROM.md`.

Review. A second AI system (gpt-6-astra via Codex) reviewed each working round and this package. It recomputed several numbers independently. Its corrections are applied (section 7). Its new numbers were rechecked with code: `verifica_numeri_astra_t1.py`, `ricontrollo_ta_dopo_astra.py`, and a root search for the complex-momentum singularities of section 4.6. No human peer review is claimed. The literature check covers the works listed in section 2, not the whole literature.

Working folders. In the working records the material is in folders named 01_mappa (map, premises and their readings), 02_fermioni, 03_U1 and 04_gruppo_record. The working files are in Italian.

Where the programs of each section are.

- 4.1 charges (P2a–P2c): `codice/03_U1` (`tappa2_carica.py`, `ricontrollo_dopo_tappa2.py`).
- 4.2 three readings (T-A): `codice/01_premesse` (`ta_tre_letture.py`; the recheck after the review is `ricontrollo_ta_dopo_astra.py`).
- 4.4 gluons (G1–G3): `codice/04_gruppo_record/gluoni.py` and `ricontrollo_dopo_gluoni.py`.
- 4.5 electroweak bosons (E1–E4): `codice/04_gruppo_record/debole_due_unita.py` and `ricontrollo_dopo_debole.py`.
- 4.6 one hand (O1–O5, H1–H2): `codice/02_fermioni/overlap_240.py`, `o5_h2_rapido.py`, `collina_h1.py` (with `NOTE_collina_h1.txt`: a check announced in its docstring is not executed). In the index convention, `overlap_240.py` counts the exact zero modes as $n_- - n_+$; `o5_h2_rapido.py` uses the spectral asymmetry.
- 4.6 pion: `pione_elicita.py` (the earlier working note, whose reading is corrected in the text) and `pione_elicita_v2.py` (the corrected version), in `codice/01_premesse`.
- 4.7 winding (T1–T6): `codice/02_fermioni/tappa1_*.py` (`tappa1_avvolgimento.py`, `tappa1_dispersione.py`), with `ricontrollo_dopo_tappa1.py`, `verifica_numeri_astra_t1.py`, `ricontrollo_periodo_b.py`.
- 4.9 SU(2) from the records: registrations and outcome note in `documenti/`, without programs.

How to redo the calculations.

```
python codice/run_all.py            # full run
python codice/run_all.py --short    # skips the three long programs
```

Python 3 with numpy, scipy and sympy is needed. The reference environment is: Python 3.13.13, numpy 2.4.6, scipy 1.18.0, sympy 1.14.0, Windows 11 on x86-64 (4 CPUs).

- `run_all.py` runs the included numerical programs. It compares their outputs with the reference files in `risultati/`. It is a strict reproduction of the working logs, not by itself a scientific verification. It does not reproduce T-D, Tappa 3, the comparisons with published data or the bibliography checks.
- Decimal numbers must agree within $|u - v| \le 1e\text{-}9 \cdot \max(|u|, |v|) + 1e\text{-}12$; integers exactly. Quantities obtained by subtracting close numbers, and the text "np.float64(...)" printed by numpy 2, may need the reference environment to match.
- `overlap_240.py` stops after its first O5 case, as in the working run. The comparison covers O1–O4 and that case.
- One line of `verifica_numeri_astra_t1.log` was added by hand in the working records after the run. It is excluded from the comparison. Its claim about periodicity is checked separately by `ricontrollo_periodo_b.py`, written for this deposit. The check is exact for a constant mass term, as there; it is not extended to the Wilson term that depends on k.
- `--short` skips `tappa1_avvolgimento.py`, `overlap_240.py` and `o5_h2_rapido.py`. The Tappa 1 recheck then uses the reference `tappa1.json` as input and does not reproduce its generation.
- Before the deposit the full run was made from scratch. The output and the running times are in `documenti/COLLAUDO_COMPLETO.txt`.

Contents of the package. Besides the programs listed above there are: `risultati/` (reference outputs of the working runs); `documenti/REGISTERED_TESTS.md` (every registered prediction, abridged in English, with its outcome); `documenti/registrations_original/` (the registered files in Italian and the registration logs, `REGISTRO.txt`: UTC time, SHA-256, file name; the logs also list working files that are not in this deposit); `documenti/working_notes_it/` (outcome notes in Italian, including those written after each review); `WHERE_THIS_COMES_FROM.md` (optional background); `LEGGIMI_IT.md` (summary in Italian); `IMPRONTE.sha256` (fingerprints of every file).

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made, then the rewriting for a school student. The numbers were checked with a program against the archived text.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **Algebra, group.** Two mathematical ways of describing symmetries. The group contains the transformations; the

algebra their "very small" versions. Algebra, group and force must be kept apart.

- **Anomaly.** The breaking, in the passage to quantum physics, of a symmetry of classical physics. If it hits the symmetries of the forces (gauge symmetries) it makes the theory inconsistent: there it must be absent.
- **Boson.** A particle with integer spin. The photon, the gluons, W and Z carry the forces; the Higgs is also a boson, but it does not carry a force as they do.
- **Chirality, hand.** Left or right, like the two hands: equal but mirror images.
- **Fermion.** A particle of matter: quarks and leptons.
- **Hill.** The Wilson term $r\,\Sigma(1-\cos k_\mu)$. It is zero at $k = 0$ (the valley) and has ridges $2r$ at the corners with n components equal to π.
- **Index.** The difference between left and right zero modes of an operator.
- **Lattice.** A grid of points in space, like the crossings on a sheet of squared paper.
- **Link.** The segment between two neighbouring points of the lattice.
- **Link transport.** The matrix that carries the internal components of a field from one point of the lattice to the next: a phase for light, 2×2 for the weak force, 3×3 for colour.
- **Overlap.** A construction on the lattice that keeps only one light particle, with the twins pushed to the scale of the cell, and that preserves on the lattice the symmetry between the hands in the Ginsparg–Wilson form.
- **Plaquette.** The small square of the lattice; the plaquette action adds up one contribution for each small square.
- **Premise.** A starting point of the author, like an axiom.
- **Reading.** In MM, a registered observation. Here, the way in which a force acts on a field: on which components and on which hand.
- **Reading level m_0.** The mass parameter of the overlap. The light species are those with Wilson term below m_0.
- **Registered.** Written in a file and given a fingerprint before the calculation.
- **Standard Model.** The theory of the particles and of the strong, weak and electromagnetic forces.
- **Tree level.** The first level of calculation, the simplest one, without virtual particles.
- **Twin, doubler.** An extra light species, forced by the lattice (Nielsen–Ninomiya), like an echo of the real particle.
- **Winding W.** The number of times a map wraps around its target, like a rubber band around a finger. Technically: the three-dimensional winding number of a map from the Brillouin torus to SU(2) or U(n). For $SU(2) \cong S^3$ it is the degree of the map. When $\pm I$ are regular values it is equal to the sum of the node chiralities, defined from the local effective Hamiltonian, at quasi-energy 0 and, separately, at π.

References

Works cited in the record, with their DOI.

- A et al. (2000). Hierarchies without symmetries from extra dimensions. https://doi.org/10.1103/physrevd.61.033005
- Bagdonaite et al. (2013). Robust Constraint on a Drifting Proton-to-Electron Mass Ratio at z=0.89 from Methanol Observation at Three Radio Telescopes. https://doi.org/10.1103/physrevlett.111.231101
- Bisio et al. (2014). Quantum field as a quantum cellular automaton: The Dirac free evolution in one dimension. https://doi.org/10.1016/j.aop.2014.12.016
- Burns et al. (2024). Constraints on variation of the weak scale from big bang nucleosynthesis. https://doi.org/10.1103/physrevd.109.123506
- D’Ariano et al. (2014). Derivation of the Dirac equation from principles of information processing. https://doi.org/10.1103/physreva.90.062106
- Furey (2014). Generations: three prints, in colour. https://doi.org/10.1007/jhep10(2014)046
- Furey (2016). Charge quantization from a number operator. https://doi.org/10.48550/arxiv.1603.04078
- Ginsparg et al. (1982). A remnant of chiral symmetry on the lattice. https://doi.org/10.1103/physrevd.25.2649
- Higashikawa et al. (2019). Floquet Chiral Magnetic Effect. https://doi.org/10.1103/physrevlett.123.066403
- Kadoh et al. (2007). A construction of the Glashow-Weinberg-Salam model on the lattice with exact gauge invariance. https://doi.org/10.48550/arxiv.0709.3658
- Kaplan (2011). Chiral symmetry and lattice fermions. https://doi.org/10.1093/acprof:oso/9780199691609.003.0004
- Lüscher (2002). Chiral gauge theories revisited. https://doi.org/10.1142/9789812778253_0002
- Navas et al. (2024). Review of Particle Physics. https://doi.org/10.1103/physrevd.110.030001
- Nielsen et al. (1981). Absence of neutrinos on a lattice. https://doi.org/10.1016/0550-3213(81)90361-8
- Pati et al. (1974). Lepton number as the fourth "color". https://doi.org/10.1103/physrevd.10.275
- Shaposhnikov et al. (2009). Asymptotic safety of gravity and the Higgs boson mass. https://doi.org/10.1016/j.physletb.2009.12.022
- Sun et al. (2018). Three-Dimensional Chiral Lattice Fermion in Floquet Systems. https://doi.org/10.1103/physrevlett.121.196401
- Uzan (2011). Varying Constants, Gravitation and Cosmology. https://doi.org/10.12942/lrr-2011-2
- Wilson (1974). Confinement of quarks. https://doi.org/10.1103/physrevd.10.2445

- Yoo et al. (2003). Big bang nucleosynthesis and cosmic microwave background constraints on the time variation of the Higgs vacuum expectation value. https://doi.org/10.1103/physrevd.67.043517